Homework 2

Wave Digital Filter

Simulating piezoelectric MEMS loudspeaker

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Abstract

This homework addresses the modeling of a piezoelectric MEMS loudspeaker using the Wave Digital Filter (WDF) technique. Starting from a linear lumped-element electro-mechano-acoustic model, the system is reformulated into the mechanical domain to simplify the circuit structure and facilitate the WDF implementation.

Contents

1 Introduction

This report focuses on the development of a Wave Digital Filter (WDF) model for a piezoelectric MEMS loudspeaker, based on the linear lumped-element reference circuit provided in the assignment. The original analog model is converted into its WDF equivalent using a fully explicit computational approach, deliberately excluding iterative solution methods. The performance of the implemented WDF model is then evaluated by comparing its output to the reference signal obtained from a Simscape simulation, considering both time-domain and frequency-domain analyses.

2 General WDF Scheme

The WDF tree structure designed for this model is illustrated in Figure 1. The input voltage source $F_{in}(t) = \alpha V_{in}(t)$ is selected as the root of the tree, as it cannot be adapted within the WDF framework. The remaining components are systematically arranged in a binary tree configuration, incorporating both series and parallel adaptors. Each adaptor is adapted by appropriately selecting impedance values, as detailed in the following computations.

All adaptor ports are numbered and connected to the corresponding wave digital elements to ensure correct signal flow.

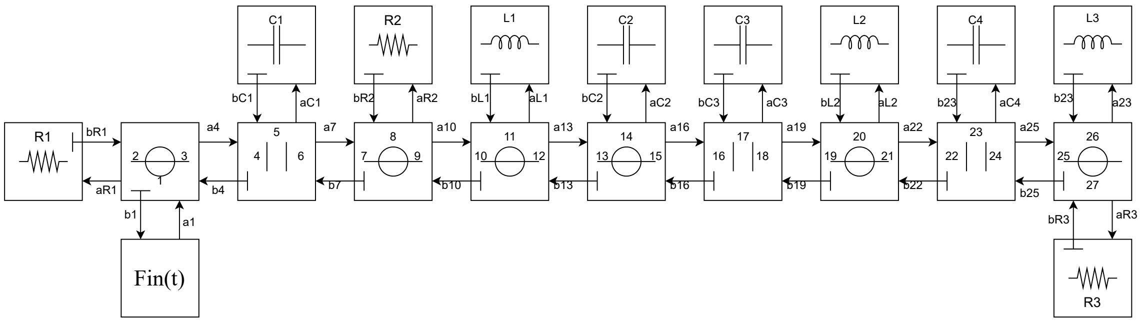

Figure 1: WDF topology representation

3 Adaptation

In the WDF structure, it is necessary to assign a free parameter Z_k to each port to enable correct wave variable transmission throughout the system. These parameters are directly derived from the characteristics of the physical components and their corresponding wave domain representations. The sampling period is defined as $T_s = \frac{1}{f_s}$, where the sampling frequency is f_s set to 192 kHz. Each circuit element is matched with its adapted impedance according to its nature: $Z_R = R$ for resistors, $Z_C = \frac{T_s}{2C}$ for capacitors, and $Z_L = \frac{2L}{T_s}$ for inductors. The component values used here refer to the previously converted mechanical-domain model, in which the ideal transformers were eliminated. The impedances assigned to resistive, capacitive, and inductive elements follow directly from this domain transformation.

$$
\begin{aligned}
&Z_{R1} = R_1, \qquad Z_{R2} = R_2, \qquad Z_{R3} = R_3 \\
&Z_{C1} = \frac{T_s}{2C_1}, \qquad Z_{C2} = \frac{T_s}{2C_2}, \qquad Z_{C3} = \frac{T_s}{2C_3}, \qquad Z_{C4} = \frac{T_s}{2C_4} \\
&Z_{L1} = \frac{2L_1}{T_s}, \qquad Z_{L2} = \frac{2L_2}{T_s}, \qquad Z_{L3} = \frac{2L_3}{T_s}
\end{aligned}
$$

Each impedance was then systematically labeled and assigned to its corresponding port within the WDF tree. The following list details the port-to-impedance mapping:

$$
\begin{aligned}
&Z_{27} = Z_{R3}, \qquad Z_{26} = Z_{L3}, \qquad Z_{23} = Z_{C4}, \qquad Z_{20} = Z_{L2}, \\
&Z_{17} = Z_{C3}, \qquad Z_{14} = Z_{C2}, \qquad Z_{11} = Z_{L1}, \qquad Z_8 = Z_{R2}, \\
&Z_5 = Z_{C1}, \qquad Z_2 = Z_{R1}
\end{aligned}
$$

To guarantee reflection-free operation throughout the WDF network, impedance adaptation was applied recursively following the appropriate series and parallel combination rules. The complete set of adaptation conditions is provided in the following section.

3.1 Series:

$$
\begin{aligned}
&Z_{25} = Z_{26} + Z_{27}, \qquad Z_{19} = Z_{21} + Z_{20}, \qquad Z_{13} = Z_{15} + Z_{14}, \\
&Z_{10} = Z_{11} + Z_{12}, \qquad Z_7 = Z_8 + Z_9, \qquad Z_1 = Z_3 + Z_2
\end{aligned}
$$

3.2 Parallel:

$$
Z_{22} = \frac{Z_{23} \cdot Z_{24}}{Z_{23} + Z_{24}}, \qquad Z_{16} = \frac{Z_{17} \cdot Z_{18}}{Z_{17} + Z_{18}}, \qquad Z_4 = \frac{Z_5 \cdot Z_6}{Z_5 + Z_6}
$$

3.3 Equality Assignments:

$$
\begin{aligned}
&Z_{24} = Z_{25}, \qquad Z_{21} = Z_{22}, \qquad Z_{18} = Z_{19}, \qquad Z_{15} = Z_{16}, \qquad Z_{12} = Z_{13}, \qquad Z_9 = Z_{10}, \\
&Z_6 = Z_7, \qquad Z_3 = Z_4
\end{aligned}
$$

3.4 Scattering Matrices:

Next, the scattering matrices S for each adaptor were calculated to accurately represent wave propagation within the structure. For a series junction, the scattering matrix is defined as:

$$
S = I - 2ZB^\top (BZB^\top)^{-1} B
$$

,

$$
S = 2Q^\top (QZ^{-1}Q^\top)^{-1} QZ^{-1} - I
$$

3.5 Wave Computation:

At every port, two wave variables are defined: the incident wave $a[k]$, which is updated during the backward traversal of the WDF tree, and the reflected wave $b[k]$, which is calculated recursively based on the scattering relations at each junction. For memory elements such as capacitors and inductors, the reflected wave depends on the value of the incident wave from the previous time step:

$$b[k] = a[k-1] \text{ for capacitors}$$

$$b[k] = -a[k-1] \text{ for inductors}$$

These recursive update rules are essential to correctly capture the time-dependent behavior of energy-storing components in the discrete-time domain.

4 Results:

Figure 2 presents a comparison between the output signal $p_{out}(t)$ generated by the WDF model and the reference signal from the ground truth, Figure 3 presents a Time-domain comparison and Figure 4 the corresponding error signal.

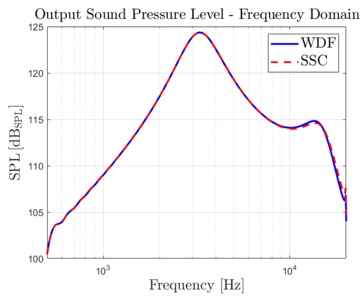

Figure 2:

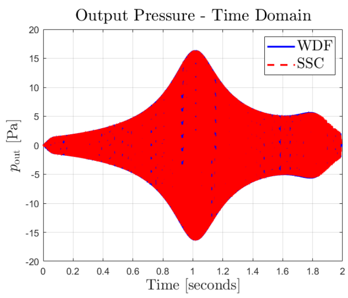

Figure 3:

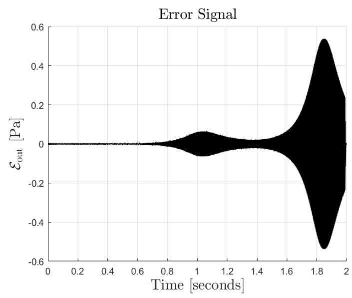

Figure 4:

The output shows a strong agreement with the ground truth, with the Mean Squared Error (MSE) calculated as:

$$MSE = 1.19 \times 10^{-2}$$